Gamma-ray Analysis for Multi-Messenger Astrophysics: GAMMA

Joan Alcaide-Núñez[1] and Martina Solano Soto[2]
[1] *Ludwig-Maximilian-Universität München (LMU München), Munich, Germany*
[2] *affil*
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Abstract.

CONTENTS

I. INTRODUCTION

The Gamma-ray Analysis for Multi-Messenger Astrophysics (GAMMA) initiative is a project by the CAPIBARA Collaboration with the goal to study the high-energy, gamma-ray phenomena in the universe and launch a CubeSat to observe these events. In this white paper, we provide a set of compelling open problems that motivate the development and launch of such a mission.

In the 2020 decadal survey [**?**], it was pointed out that multi-band, that means acros the electromagnetic spectrum, and multi-messenger, that means combining different "messengers" such as light, GW and neutrino, detections were to be prioritized in the next years. Currently operating flagship gamma-ray observatories have long passed their main operation lifetimes, and new replacements are still in very early phases.

Professionally or student developed CubeSats with miniaturized detector technology have proven to be able to successfully detect gamma-ray bursts (GRBs) [**? ? ? ?**]. However, gamma-ray photons cannot be focused as with a traditional optical telescope. A good way around to accurately observing and locating GRBs is to perform triangulation with multiple satellites in orbit. Therefore, having multiple missions with large fields of view and good resolution capabilities is of very much importance for the proper study of this astrophysical phenomena.

Missing clear explanation of what are gamma-ray bursts and why we need to send detectors to orbit?

II. MULTI-MESSENGER ASTRONOMY

As previously discussed, multi-messenger astronomy is a new way of observing the universe by combining different types of information, these are primarily electromagnetic waves, cosmic rays, neutrinos and gravitational waves. This section covers the fundamental questions about the topic and then proceeds to show how a student led mission can work towards the new way of understanding the cosmos. First of all, a clarification about these "messengers" is needed: they are different ways of measuring and observing cosmic events. The most historically used is light, which is a spectrum of electromagnetic waves going from radio waves all the way to gamma rays; due to its widely spread range of wavelength, it has to be divided in smaller bands and observed with wavelength specific telescopes. To get a deeper understanding, astronomers commonly combine different bands together, with the clearest example being the production of images through the combination of different photos taken in various bands (multi-band imaging).

Recently researchers have switched their focus towards a more complementary approach, in fact, the universe does not only emit light but also provides information through other probes; cosmic rays are one of them, charged particles with relativistic speed coming from our sun primarily but also from extreme astrophysical phenomena happening everywhere in space. Neutrinos are nearly massless particles which travel through the Universe with minimal interaction, providing unique insights of their sources which are supernovae, black holes, AGN etc. Finally there are gravitational waves: produced by the motion of masses through the fabric of space-time, the ones produced by super-dense objects can be detected by highly-sensitive interferometers. The majority of these messengers are connected to supernovae explosions, fusion of black holes or neutron stars and overall extreme energy conditions.

Multi-messenger astronomy was born on 17 August 2017 when a gravitational wave detection was followed by a gamma ray observation [**?**] (should I explain more in detail what was found and how they detected GW?) From that moment on, the scientific community's goal was to combine different types of signals to get a better picture of what space really looks like; a common analogy is that of a garden: sight alone is not enough to fully experience it — one also needs to touch, smell, and hear it. In the span of 9 years, only a few multi-messenger observations have been confirmed, but a huge amount of raw data, produced by the interferometers LIGO, VIRGO and KAGRA and neutrino detectors, are being analyzed. Neutrino and photonic observations have been useful to find blazars [?] for example.

However there is still a long way to achieve efficient and daily observations and in the years to come the scientific community expects to intensify research and results thanks to the upcoming Einstein Telescope, LISA and IceCube-Gen2. GAMMA contributes to this shared goal through its gamma-ray observations. More specifically, GAMMA fills the gap left by existing instruments, such as FERMI launched in 2008, which are decades old and have passed their operational lifetime; these telescopes don't fully cover the gamma ray frequency either, missing crucial information which our mission plans to acquire. The choice of GAMMA comes from this specific lack of instruments alongside the role that the gamma band plays in the multi-messenger era: gamma rays are the most energetic ones, thus we expect to observe them together with extreme events, that's exactly what happened with GW170817, and for this reason a follow up observation in this band rather than the others is very impactfull.

In conclusion, multimessenger astronomy opens new frontiers but requires more effort and instrumentation; a student-led mission can be as flexible as it needs while also keeping the right amount of rigor to be part of the complex scientific research.

III. COSMOLOGY: GRBS AS HIGH-z PROBES

- intro to cosmology: the LambdaCDM model as the model of the universe (explaining accelerated expansion from EFE)
- explain the problem of the Hubble tension (what is wrong with our current standard model of the universe?)
- how GRB can be used to probe the high-z redshift, why does it matter to probe high-z and what do we need in terms of instrumentation

IV. LORENTZ INVARIANCE AND HIGH-ENERGY PHOTON LAGS

- briefly explain why a theory of quantum gravity is of such relevance
- elaborate on how detecting high-energy photon lags provides direct proof of a granular space-time
- transform theory into timing requirements

V. SUMMARY OF TECHNICAL REQUIREMENTS

VI. CONCLUSIONS

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